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To cite this article: Liza Bolitzer (07 May 2026): Analytic questions: asking, listening, and responding in data analysis, International Journal of Social Research Methodology, DOI: [10.1080/13645579.2026.2660843](https://doi.org/10.1080/13645579.2026.2660843)

To link to this article: <https://doi.org/10.1080/13645579.2026.2660843>



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Published online: 07 May 2026.



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RESEARCH ARTICLE

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Analytic questions: asking, listening, and responding in data analysis

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ABSTRACT

This methodological paper presents the use of analytic questions (AQs) in qualitative data analysis. AQs are a way to identify and analyze data so a researcher may respond to a study's central research questions (RQs) or adapt RQs to emerging themes and/or findings. AQs can move successively upward from looking at individual participants or cases to the study sample to larger-scale statements of insight. This paper presents the use of AQs in a qualitative study of adjunct faculty to explore the process of employing analytic questioning in data analysis. Through a detailed exploration of AQs, this paper aims to guide qualitative researchers in employing AQs in their own studies and further the transparent description of data analysis within presentations of qualitative research.

ARTICLE HISTORY

Received 2 December 2024
Accepted 11 April 2026

KEYWORDS

Data analysis; analytic questions; qualitative research; Iterative Analysis

Qualitative research has a strong tradition of using questions as a guide. Well-articulated research questions (RQs) can focus researchers on the topic under investigation and enable them to clearly express how they are approaching that topic (Bryman, 2007; Denzin & Lincoln, 1994; LeCompte & Schensul, 2010; Marshall et al., 2021; Maxwell, 2013). During the process of designing and implementing a study, RQs function as a central reference point, as Luttrell (2010) describes, '[a] "north star" in that it orients the researcher to where they are and what direction they are traveling' (p. 6). As studies move from initial conceptualization to data collection, to analysis, and finally to presentation, qualitative researchers will likely return to and reform their RQs (Agee, 2009; Lareau, 2021). To extend Luttrell's metaphor, as a study progresses, researchers may need to identify a new 'star'; new or altered RQs they seek to answer. This process – identifying, forming and reforming RQs in the research process – is well-recognized among scholars of qualitative research (Agee, 2009; Bazeley, 2013; Creswell & Poth, 2018; Lareau, 2021; Luttrell, 2010). Less discussed, however, is how such a process may unfold in data analysis.

Although the importance of RQs is commonly discussed in research methods texts, with whole chapters often dedicated to them (e.g. see Bazeley, 2013; Creswell & Poth, 2018), less explored in these texts is the process of forming and reforming questions in the data analysis process. Just as RQs can direct and guide a study, identifying and posing questions during the data analysis process can enable researchers to discover and analyze

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patterns in study data (Bazeley, 2013; Bingham et al., 2016; Neumann & Pallas, 2015). In forming and posing analytic questions (AQs) to study data, researchers are forced to articulate what they are trying to understand in the data analysis process, thereby enacting an essential part of qualitative research: iterative, recursive analysis (Maxwell, 2013). Yet, despite this process likely occurring in the work of many qualitative researchers, I have rarely seen it described in presentations of research studies (for exceptions, see: Bingham, 2017; Neumann, 2006).

The lack of naming the use of AQs, likely limits the transparency and, therefore, the trustworthiness of qualitative research. As Bazeley (2013) states,

Transparency involves clear explanation of process that allows the reader to understand how you, as researcher, progressed from initial purposes, assumptions, and questions through data analysis to the results, interpretation, and conclusions of your study. (p. 407)

Transparency within the presentations and publications of research is a necessary condition from which scholars can assess the validity or trustworthiness of a study (Cho & Trent, 2006; Shenton, 2004). (Guba 1981), Without transparency, we are unable to understand and evaluate the methods researchers use to work towards the conception of validity or trustworthiness that they are employing (Marshall et al., 2021). Furthermore, as Maxwell (2013) argues, validity or trustworthiness is not achieved simply through the employment of methods:

The evaluation of [...] conclusions depend not simply on the appropriate use of the procedures, but on what the data generated by these procedures allow the researcher to conclude. (p. 132)

Said another way, it is not enough for us to state what we do, we need to show the relation of the steps that led to the research findings. In this paper, I offer AQs as a way to both show and enact those analytical, relational steps in the data analysis process.

In sharing this process of using AQs in this paper, I have two goals: one, to show the process of using AQs to researchers who are unaware of this approach because it is largely unnamed; and two, to provide a name and rationale for the use of AQs those who are already intuitively engaging in this process but do not yet have the vocabulary to describe their actions.

The first goal focuses on teaching about AQs, while the second – naming AQs – focuses on making what may be an existing process more explicit. The value of the first goal is straightforward. I am presenting readers with how to incorporate questioning into analytic processes, which is a recognized goal of qualitative researchers (Bazeley, 2013; Bingham et al., 2016; Neumann & Pallas, 2015). The second goal – naming what may be an existing, intuitive process – is more complex, as it rests on three assumptions.

One, I am assuming that intuitive data analysis processes can be made more explicit without diminishing their value (Sybing, 2022) – that by naming AQs we do not limit intuitive processes of questioning within data analysis either by AQs or by other means. My belief is the opposite, that by naming and discussing that naming process, researchers support their questioning. In making this assumption, I draw from the common maxim in qualitative research that the researcher is the instrument (Yoon & Uliassi, 2022). This metaphor captures the way in which we engage in cognitive, emotional, physical and relational processes as researchers (Lavee & Itzhakov, 2023). It also encourages us to

care for and tune ourselves as instruments by making those processes clear to ourselves and others. A key example of this is how qualitative researchers engage their subjectivity, the ‘garment that cannot be removed’ (Peshkin, 1988, p. 17), throughout the research process. As I detail in the example below, AQs can support that process.

Two, I assume that being explicit about processes of questioning within data analysis benefits both individual researchers and the larger field. Guides to data analysis often stress how audit trails provide researchers with a way to examine and strengthen their analysis, thereby increasing the rigor of the study. In sharing their audits, researchers can increase the transparency and, therefore, credibility of their work (see Bazeley, 2013; Marshall et al., 2021). Furthermore, articulating AQs and responding to them within the data analysis and collection process has increased, in my view, the depth of my findings, a key element of rigor in qualitative research. My experience is also reflected in the work of other scholars who have named this practice in their work (see: Bingham, 2017; Neumann & Pallas, 2015).

And lastly, I assume that using AQs does not preclude other analytic strategies. As I detail in the example below, I find the opposite; AQs can complement and expand on other analytic approaches. I present these three assumptions that underlie the value of naming AQs as assumptions because I view them as open to inquiry. In writing this paper, I am inviting further discussion of both the use and value of AQs.

Towards the two goals delineated above, this paper begins with a brief discussion of AQs in data analysis, including their potential use at different stages of data analysis. I then expand on that definition in the subsequent section by providing an example of the use of AQs in the analysis of study data for one research study. In presenting this example of how I used AQs, my goal is to show the data analysis process as it occurred, thereby encouraging other researchers to make their own iterative processes of questioning in data analysis transparent. In the discussion section, I offer ideas for how the use of AQs may be incorporated into other approaches to data analysis.

Analytic questions (AQs)

AQs are questions formed and reformed in data analysis that enable researchers to respond to the RQs and interrogate emerging patterns or themes (Felix, 2021; Neumann & Pallas, 2015). Unlike the larger RQs directed at the study as a whole or the larger research universe in which the study is situated (Maxwell, 2013) or interview questions directed at individual participants (Seidman, 2006), AQs are directed at specific aspects of the data: they are questions posed to the data, enabling researchers to probe and expand their understandings. As Bingham (2017) states, employing AQs is ‘a process of continually asking questions of the data in order to let the findings emerge’ (p. 531). It is important to note, however, that this emergence is not an open, strictly inductive process, such as within open-coding (see: Cascio et al., 2019). As detailed below, in initially forming and using AQs, the researcher is directing their analytic eye, formed through their RQs and conceptual framework, toward specific study data. As Neumann and Pallas (2015) state, AQs are ‘like a small shovel, shaped (and iteratively shaped) to enable the researcher to “scoop” out of the data a subset that can respond to a question of interest’ (p. 166). Using this metaphor, the shape of the shovel guides the researcher toward what is ‘scooped’ (i.e. the data that fit in the shovel). The emergence, therefore,

occurs through the careful crafting and use of the shovel. However, as Neumann and Pallas (2015) state, and I detail in this paper, this is also not strictly a deductive process. The shaping of the AQs is an iterative process, whereby the researcher moves between forming their shovel, reflecting on what it enabled them to scoop, and reforming their shovel to probe the data further. That reforming can often include inductive processes whereby new AQs emerge from the data itself. As such, AQs can serve three interrelated functions in data analysis: One, they can enable researchers to identify data that responds to RQs; Two, they can be formed to probe additional topics or themes that emerge in data analysis; and, three, they can enable the researcher to interrogate relationships within subsets of the data. I detail each of these functions below.

One, AQs can direct the researcher to certain parts or aspects of data that respond to the RQs. For example, an AQ may ask, ‘What does this study data tell me about XYZ?’ when ‘XYZ’ has been identified in the study’s RQs or conceptual framework. Posing this question will direct the researcher to pull data that speaks to ‘XYZ’ from particular study data (e.g. a single interview or document) and then respond to the question by writing an analytic memo or by creating a data display. In this function, AQs help the researcher to think deeply in the coding process about how pieces of study data connect, counter, or expand on the RQs and the conceptual framework guiding the study. This first function is often the initial way AQs are used in data analysis.. As detailed below, as the researcher moves more deeply into the study data, AQs can form more directly from the data itself, helping the researcher to identify patterns not anticipated during initial study design.

The second potential function of AQs is to help the researcher identify additional topics or themes that emerged in data analysis. For example, the researcher may note the repetition of a phrase or idea in the study data and then pose AQs aimed at understanding that thing. These AQs can include ‘why’ questions like, ‘Why is [insert phrase or idea] so important to participants?’ where the phrase or idea is something the researcher did not anticipate prior to data collection. Or they may pose ‘how’ questions like, ‘How are participants defining [insert phrase or idea]?’ for phrases or ideas that came from participants. Posing and responding to AQs such as these can help the researcher to return to the study data overall and look at the data in a new way. These AQs can also help push the researcher beyond describing the data to identify an element or process they did not anticipate emerging in the study data. To probe that element or process, the researchers may need to approach their data in new ways, prompting them to identify additional theories or concepts outside their original conceptual framework. This can also be an opportunity for the researcher to interrogate their own positionality, the beliefs and viewpoints that they carried into the research process (Peshkin, 1988). By posing questions like, ‘How does [emerging theme] contradict my beliefs about [research topic]?’ or ‘what might be preventing me from understanding [emerging theme]?’ the researcher may be able to see both their data and themselves in new ways, thereby sparking new avenues of insight. This second function of AQs is an example of how posing AQs can support the larger, iterative process of qualitative research, where the researcher forms and reforms the focus of the study throughout the research process (Luttrell, 2010).

A third potential function of AQs is to help researchers interrogate relationships within subsets of the data (Bazeley, 2013). For example, AQs can be used to interrogate the relationship, repetition, or the absence of certain codes in the data. For example, the

researcher may ask, ‘When do participants discuss XYZ and QRS together?’ or ‘Why do these study participants discuss XYZ and others do not?’ Answering these AQs can help researchers to examine the relationships between data segments, as well as the absence of patterns in some participants. This can be very helpful in identifying negative cases or outliers, which is a key strategy in working towards trustworthiness in qualitative research (Bazeley, 2013; LeCompte & Schensul, 2013). Furthermore, if data analysis is occurring within data collection, this third function of AQs may prompt the researcher to return to their participants with new questions aimed at better understanding the relationships within the study data.

Within these three functions, AQs can move successively upward from looking at individual participants or cases to the study sample to larger scale statements of insight (Neumann & Pallas, 2015). Researchers may employ them initially to identify data, reform them to probe that data across participants, and form new AQs to interrogate relationships between data or themes further or to identify what is missing. In the example I present below, I demonstrate this process of moving successfully upward. However, it is important to note that these functions may also be applied separately or in other orders, depending on the research approach. For example, a researcher engaged in a phenomenological inquiry might begin data analysis by coding their data with deductive codes based on a conceptual framework that encapsulates theories and research aimed at understanding the phenomenon they are studying (Bazeley, 2013). Having done so, they might then choose to focus on the relationship between codes, the second function detailed above. Or, in that initial deductive coding they may notice the absence of the application of a code and their next step may be to probe that absence through posing new AQs, which can lead to additional data collection. Another researcher, working from a more ethnographic, grounded approach, might begin their analytic process by deeply immersing themselves in their data, reading and re-reading transcripts as they write memos or reflect on what they see in their data (LeCompte & Schensul, 2013). From that immersive process, they may form initial analytic questions of the second function, which allow them to probe what is emerging from the data itself. How a researcher uses AQs will depend, therefore, on their overall approach to qualitative research.

In addition to the variation in how researchers employ AQs, researchers can also respond to their AQs in multiple ways. For example, they may write analytic memos where they draw on the study data to write responses to the questions (Bazeley, 2013). Or they might create data displays or matrixes (Miles et al., 2020) which provide a visual representation of how the data respond to the AQ. The decision of how to respond to the questions depends on both what the researcher is trying to understand about their data and their research approach. For example, a researcher engaging in content analysis who is deeply analyzing word usage is likely to create matrices that can show placement and frequency, while a researcher engaging in a narrative inquiry is more likely to write memos in which they capture and analyze the stories of their participants (Bazeley, 2013). Furthermore, as researchers respond to these questions, they may identify new questions that take their analysis in new directions, prompting them to pose additional AQs which they respond to in additional ways.

It is important to note that none of what I have described above is revolutionary in qualitative data analysis. The use of AQs works in tandem with the well-established

traditions of coding, memo-ing, and carefully probing study data (Bazeley, 2013; LeCompte & Schensul, 2010; Marshall et al., 2021; Maxwell, 2013), but it also offers two additional pieces: One, it presents three key ways of forming and reforming questions in data analysis; and, two, for qualitative researchers who are already engaging in this process of forming questions, it provides a name for what they are doing so that they might more readily include that process in their descriptions of data analysis. Returning to the metaphor of AQs as scooping, this paper aims to support researchers in both forming their shovels and being clearer about the nature of the shovel they have formed. Given the current, limited discussion of AQs in research methods texts, I am unable to know how novel this approach will be to readers.

In the next section, I provide an example of the use of AQs. In this example, AQs are the central analytic strategy, employed across the data analysis process in concert with other analytic tools (coding, memo'ing). However, as detailed above, researchers may also use AQs as a more discreet tool within another analytic framework. The example below is, therefore, offered as one example of how researchers may use AQs and not as a strict model for their usage.

Example study: adjunct faculty as teachers

To ground my description of the use of AQs in data analysis, I use as an example a study I conducted on adjunct faculty as teachers (Bolitzer, 2019a, 2021). This study examined how non-tenure track, adjunct instructors of general education classes learned to teach and what they came to believe about teaching 'on the job', in their day-to-day work as instructors. I approached this study from a constructivist perspective, which views people's beliefs and actions as continually negotiated with others in particular contexts that draw out particular meanings (LeCompte & Schensul, 2010). My aim was to learn from participants about their teaching beliefs and to see how they sought to apply those beliefs in their classrooms (Bolitzer, 2019a, 2021). To demonstrate how I used AQs in the data analysis process for this study, I begin by briefly summarizing the study's conceptual framework, RQs, and data collection process. This background is necessary because these elements of the study informed my use of AQs. I provide a much larger explanation of these elements in other works (see: Bolitzer, 2019a, 2021), but because the aim of this paper is to focus on the use of AQs, here I am brief, only presenting the key components that relate most directly to data analysis.

Conceptual framework

This study is anchored in research on adjunct faculty as teachers (Bolitzer, 2019b; Gappa & Leslie, 1993; Kezar & Sam, 2012) and theories of teachers' professional learning (Bourdieu & Passeron, 1990; Hansen, 2001; Shulman, 2004; Wenger, 1998). Central to this framework is the idea of *learning within teaching*, defined as a teacher's ongoing engagement with their students; subject matter(s); and the various contexts (e.g. social, academic, economic, political) with which the teacher, student, and subject matter interact (Ladson-Billings, 2006; Shulman, 2004). The comingling of these elements may

result in new or altered understandings for teachers about teaching, which they may then apply in their work (Schwab, 1983).

This conceptual framework guided me in focusing the study on what adjuncts learned within the work of teaching itself – as they planned for teaching, engaged with students in the classroom, and reflected on those exchanges. The framework also helped me to think about what prior knowledge and experience adjuncts might bring with their teaching and how the teaching space might become a place to form and reform their beliefs.

RQs

The following RQs (i.e. my ‘stars’) guided the initial design of the study and the data analysis process:

- (1) How do adjunct instructors of general education courses at highly diverse institutions of higher education learn to teach as they carry out their teaching responsibilities – including as they prepare to teach, enact their teaching, and possibly reflect on their teaching?
- (2) What do adjunct instructors of general education courses at highly diverse institutions of higher education learn about teaching, and about matters related to teaching, as they engage in it, including as they prepare to teach and as they reflect on their teaching?

Data collection

Data collection was an iterative process of interviews (Kvale & Brinkmann, 2009; Seidman, 2006) and classroom observation (Atkinson & Hammersley, 1997; Erikson, 1986), which enabled me access and contextualize participants’ learning within teaching (Bolitzer, 2020). Prior to data collection, I received approval for the study from the Institutional Review Board for the Protection of Human Subjects at Teachers College, Columbia University (Protocol 15–439, titled: ‘Learning within Teaching: A study of adjunct faculty’). After obtaining IRB approval, I solicited participants through their academic departments. I began by interviewing each participant for 1–2 hours to understand their history, beliefs, and experiences as teachers. I then observed their teaching 2–3 times, often including field-based interviews. After completing the observations of each participant, I interviewed them a second time to understand better what I observed. Total data collected across the 19 participants included 47 interviews and 44 classroom observations, syllabi, course documents (e.g. assignments, hand-outs, readings), and curriculum vitae.

Employing AQs in data analysis

Posing AQs in data analysis enabled me to move successfully upward, from (a) looking at participants as individual cases to (b) identifying themes across study participants to (c) making claims about those themes; and (d) lastly, forming propositions for what the study may offer more broadly.

Participant-level analysis

My initial, participant-level AQs were outgrowths of the original RQs and helped me to form my initial understanding of how and what each participant learned while teaching, based on the initial interview. After transcribing and cleaning the first interview for each participant, I asked the following questions of the data:

- (1) How does this participant describe how they learned, and/or is learning, to teach as they carry out their teaching responsibilities?
- (2) How does this participant describe what they have learned about teaching?

To organize segments of data that responded to these questions, I used the data analysis software, Dedoose. I then focused on these segments to write an initial individual analytic memo for each participant, responding to the two questions. These memos enabled me to capture participants as individuals. This example is one of the first functions of AQs – they directed me to data that responded to the RQs.

After analyzing the initial interview with study participants, I then observed their classroom teaching, which often included additional field-based interviews. I then closed data collection for the individual participants with a final interview, aimed at better understanding the relationship between their beliefs and practices as a teacher (Bolitzer, 2020). These second interviews were informed by employing the second function of AQs, the identification of emergent themes. To prepare for those interviews, I sought out data segments (from the initial interview, observations and field-based interviews) that spoke to participants' learning or the contexts in which that learning unfolded but did not fit into the narrower 'how or what of learning' posed in the RQs. These AQs emerged from the data, and were specific to individual participants. For example, a participant who was an adjunct teacher in the social sciences named Anthony Thomas (pseudonym), spoke in his initial interview in great detail about realizing his students had very different skills and capacities than he did as an undergraduate. Sociological concepts such as hierarchy and power made sense much more easily to his students, given what they observed and experienced, than they did to Thomas, who was a white man from a privileged background (see: Bolitzer, 2021). I had not anticipated this response from a participant, and reflecting on what Thomas shared sparked me to turn a question towards myself: 'What do I believe about students' knowledge? And how may my beliefs be showing up in the study?'. Memo'ing in response to these questions enabled me to engage my subjectivity and, therefore, more thoroughly understand Thomas' beliefs.

Hearing from Thomas about the differences he observed between himself and his students also prompted me to pose the following AQ: How – if at all – does Thomas apply this realization of differences between himself and his students within his work as a teacher? To answer this question, I returned to my observation notes and field notes from when I observed Thomas's classes. The AQ helped me to see Thomas' teaching in new ways. For example, in my notes I had captured multiple instances of Thomas referring to his students' explanations of power. The AQ enabled me to see this as not only a teaching strategy but also a reflection of Thomas' recognition of the capacity of his students. I brought this idea to the final interview with Thomas, which led to a rich discussion of both the teaching strategy and how it developed within his

work as a teacher. Furthermore, seeing the way perception of difference showed up in Thomas' teaching led me to ask how the noticing of difference or similarity may be present for other study participants, leading me to create a code of 'difference from students,' which I then applied to other study participants within the process of coding across the study data. I found that while other participants used different terms, they likewise made sense of their students by comparing and contrasting them with themselves as undergraduates.

In addition to allowing me to immerse myself in the data, the process of applying the first two functions of AQs to all participants yielded a mix of 71 etic and emic codes, which I then applied across the study data. These codes came from the participants as individual cases but were also informed by the conceptual framework. In the first function of the AQs, I used the research questions and conceptual framework to form the AQs. These codes can, therefore, be understood as deductive, etic codes that are drawn from the study's conceptual framework and research questions (Bazeley, 2013; Bingham, 2023). Through the second function of AQs, I identified themes that emerged from the data itself. These codes can, therefore, be understood as emic, inductive codes that emerged from the study data (LeCompte & Schensul, 2013).

Identifying themes

Having completed data collection for all participants, the next step in data analysis was to move from individual participants to identifying themes across participants. To begin to do this, I continued to utilize dedoose, a web-based data analysis software, to apply all 71 codes (described above) to every participant. This enabled me to isolate 617 text segments of text across the 19 primary participants. In applying these codes, I further refined the codes themselves, often by posing AQs to them. For example, as I applied the code of 'difference,' originally derived from Thomas, I asked, 'How do participants describe differences between themselves and their students?' In answering this question, I realized many participants toggled between describing similarities and differences with their students, drawing out complex comparisons between themselves and their students; therefore, I expanded the code to include 'similarities and differences' and created subcodes with associated AQs to capture the variety.

At this stage of identifying and analyzing themes across study participants, I was able to identify the most common codes and co-occurrences of codes. This enabled me to employ the third function of AQs, which is to interrogate the relationship or repetition of certain codes or the absence of certain codes in data. Data analysis software is incredibly helpful in describing patterns and relationships; for example, Dedoose helped me to identify three ways that all participants described learning about teaching—(a) from prior experiences in education, (b) learning from students, and (c) learning from colleagues – by identifying the dual application of codes to data segments. Data analysis software does not, however, help to analyze those co-occurrences.

To move from description to analysis, I formed new AQs for each of the three ways of learning about teaching listed above. For example, to better understand how participants described their learning about teaching from prior experiences in education, I asked, 'How do participants describe the teaching practices they experienced in their own educations?' and 'How do participants seek to apply, or not apply, the teaching practices

they experienced in their own educations within their current teaching?’ To answer these questions, I once again analyzed the coded interview data and reviewed my observations of participants’ classroom teaching and the memos associated with participants. I then wrote memos that captured the patterns and differences I saw across the study participants. In this way, the AQs helped me to move from broad description to more specific claims, often guiding me back into and through the study data.

Making claims

To dig more deeply into the themes and move toward making claims about the study data as a whole, I once again formed AQs – this time drawing from aspects of the conceptual framework that spoke to each theme, as well as what emerged from themes themselves. Continuing the example of the first theme, learning from prior experiences within education, I drew on three select parts of the conceptual framework – the sociological construction of professional practice (Bourdieu, 1990; Wenger, 1998), personal biography in teaching (Greene, 1973), and teaching as vocation (Hansen, 2001) – to form the following AQs:

- (1) How do participants describe drawing from their personal biographies within their teaching?
- (2) How do participants describe working from, or with, these aspects of their personal biographies in their teaching?
- (3) What do participants describe learning, or come to believe, about teaching from bringing their personal biographies into their teaching?

I used these questions to move from describing the data pattern to making a claim about what this pattern shows. Ultimately, I was able to make the following statements about how participants drew from their personal autobiographies:

A substantial majority of study participants (i.e., 16 of 19) reported thinking about the teachers they admired or felt close to, when they were high school students or undergraduates. Participants drew especially from memories of teachers who were caring of them, and from memories of teachers who shared their enthusiasm for their subject matter. In addition to thinking about teachers they admired, many participants (i.e., 12 of 19) said they thought about themselves as undergraduate students. Some participants described themselves as similar to their students; others saw differences between themselves and their students. In each instance, participants talked about how noticing such differences and similarities led them to seek additional ways of supporting their students.

Forming propositions

As a final step in data analysis, I sought to consider what I could say, more broadly, about how adjunct faculty may learn to teach in their work as teachers. I did so by forming propositions that presented the possibility of generalizing to theory (Bazeley, 2013; Maxwell, 2013). Whereas themes, data patterns, and claims pertain to study data (here, provided by study participants), propositions consider how the study findings may apply more broadly (here, applied to adjunct instructors of general education courses).

To form propositions, I returned to the literature on adjunct faculty as teachers and the conceptual framework. I posed new sets of questions about how the study findings lined up with or enlarged existing understandings of adjunct faculty as teachers and learning for teachers more broadly. To do so, I posed the following AQ, tuned to each claim: How might this claim enlarge on, or speak back to, the extant knowledge of how college faculty learn to teach? For example, for the theme of learning from prior experiences in education, I asked: What do we know about how other teachers – such as tenure-track faculty or K–12 teachers – similarly draw on their autobiographies of education? And, if so, is it likely the adjunct faculty learning proceeds in a similar fashion, or are there unique aspects of adjuncts' work – or work lives – that shape how they may draw on their personal biographies? The answer I found by holding up my study data to the wider research literature was that participants' experiences were not unique to adjuncts. Like most teachers, they too think closely about their teaching, and their prior experiences of education inform that thinking. What was unique about them was that they did not receive the training or support that K–12 or tenure-track faculty may receive to engage in that thinking. This difference in support has been well documented in the research literature (Bolitzer, 2019b). Posing these questions enabled me to draw connections between those two sources – the study data and research literature. Thereby, I was able to offer a new insight: given the lack of support for their learning, as compared to other teachers, it is likely that for adjuncts, reflecting on past experiences may be a central space of learning.

Through posing AQs between the study data, theories of professional learning for teachers, and research literature, and listening to the responses, I was ultimately able to offer the following proposition for how other adjunct faculty – beyond those participating in this study – may learn in teaching:

Adjunct faculty likely learn to teach by drawing on their own past experiences as students, and beliefs about teaching that they developed within those experiences. They may bring those understandings into their current work as teachers, forming altered or new beliefs within the process as they learn more about their students as learners and, therefore, themselves as teachers.

Discussion

Through this paper, I have sought to show the value of AQs within the analysis of qualitative data. Through employing AQs researchers can engage in a dialogue with their study data; we can question the data, listen to the responses, and likely pose additional questions. This process can be employed throughout the data analysis process, such as within the example above, or at key moments in data analysis. In presenting AQs within this paper, my aim was both to show their process and provide a name for their use to researchers who may already be employing them. Writing this paper also enabled me to probe my own understanding of AQs, provoking some new questions about their usage. In this discussion, I will share my responses to those questions, as an invitation to other researchers to share their own responses.

A central question invoked by this paper is does the use of AQs align with particular approaches within qualitative research? In responding to these questions, I begin with a well-recognized statement: the data analysis process for any study is tightly linked to

epistemology, research approach, and research questions. For example, decolonial researchers reject positivist approaches where the researchers bring ‘externally-imposed agendas’ (Denscombe, 2024, p. 233) and, therefore, adopt participatory approaches. Their epistemology and the corresponding research approach require participant engagement across the research process, including data analysis. They may, therefore, choose to bring participants into the process of forming and posing AQs. A researcher engaging in a phenomenological study would likely take a different approach. The relationship between epistemology, research approach, and research questions requires the data analysis process to be different for distinct forms of research; a trustworthy strategy within discourse analysis is likely inappropriate to a narrative inquiry. And yet, in all forms of qualitative inquiry our goal is to move from description to analysis to findings, and potentially to larger propositions. In doing so, we are all, fundamentally, seeking to answer questions. As such, I see the potential for AQs to be employed in multiple forms of qualitative research. I also suspect they are already being employed, but researchers are not naming them as such. I hope this paper will provoke further examples of how that occurs or should occur within specific forms of inquiry. These need to come, however, from researchers operating within those traditions.

Related to the question of research approaches and AQs, there is also the question of whether AQs are deductive or inductive. As detailed in this paper, I see AQs as potentially helpful both within deductive and inductive approaches to data analysis and are, therefore, not strictly tied to a particular worldview or epistemology (Chamberlain, 2015; Pascale, 2011). As in the example above, AQs based on the conceptual framework may be used to identify text segments deductively, but then those same text segments may be probed inductively. As such, I would argue that they are not fundamentally either inductive or deductive. I also join other researchers (see: Bazeley, 2013; Saldana, 2021) in cautioning qualitative researchers to not adhere so closely to inductive or deductive approaches, which may limit their analytic processes. However, it is important to note that this caution is tied to the constructivist worldview that I bring to my research: I believe that people are actively constructing meaning in their lives in conversation with the people and society that surround them. As a researcher I, therefore, design data collection and analysis processes which will enable me to identify and understand participants beliefs from within their worlds. Prior research and theory, collected as a conceptual framework, is an essential tool for that looking. But so is immersing myself in the research process so that their beliefs can emerge from the data itself. As detailed in the example above, AQ helped me in this iterative process of moving between data collection and analysis.

The final, central question that writing this paper invoked is if AQs are so distinct from other approaches or strategies in data analysis. In describing AQs and providing an example of their usage, I have emphasized how they operate in conjunction with other data analysis strategies. For example, I use AQs as a facilitator of the coding process. Furthermore, posing AQs is only helpful if researchers respond to them and the forms of those responses draw from other, well-established analytic tools. As such, I would argue that AQs are not so distinct from other approaches or strategies. However, unlike coding or memo’ing, they are rarely named. A key reason I have written this paper is to provide that name. I hope doing so will provoke other researchers to share how they incorporate them into other approaches or strategies in data analysis.

Conclusion

As detailed in this paper, AQs helped me to form study findings that reflected the study data and to form propositions that may apply to larger populations. In this paper, I aimed to show how, in conjunction with much more recognized data analysis processes (e.g. coding, writing memos, and creating visual displays), AQs can be a valuable tool for data analysis. My hope is that through the detailed example in this paper, other researchers may be better positioned to employ AQs in analyzing their own qualitative data. I also suspect many other qualitative researchers are already employing this process but do not have the vocabulary to name it. I hope this paper will encourage them to use the term analytic questions, and to describe the process of their use.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

Notes on contributor

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